A. Proofs

A.1. Marginalization Consistency up to the Leaves of CircuITS

Lemma A.1. *A probabilistic circuit (PC) is marginalization consistent estimator up to the leaf nodes:*

$$\int p(\mathbf{y} \mid \mathcal{Q}, \mathcal{X}) d\mathbf{y}_c = p(\mathbf{y}_{-c} \mid \mathcal{Q}_{-c}, \mathcal{X}),$$

where $\mathbf{y}_{-c} = \mathbf{y} \setminus \mathbf{y}_c$*, if it fulfills:*

- ***Property 1:*** *Each leaf node in the PC represents the distribution of variables corresponding to a single input channel*
- ***Property 2:*** *The marginal leaf distributions only depend on the query subset ($\mathcal{Q}_c$) that corresponds to the respective channel*
- ***Property 3:*** *The weights of the sum nodes do not depend on the query set $\mathcal{Q}$*

Proof of Lemma A.1. Let n be a node in the Probabilistic Circuit representing the distribution $p_n(\mathbf{y}_{sc(n)} \mid \mathcal{Q}, \mathcal{X})$. We want to prove that for any channel $y_c \subseteq sc(n)$:

$$\int p_n(\mathbf{y}_{sc(n)} \mid \mathcal{Q}, \mathcal{X})\, d\mathbf{y}_c = p_n(\mathbf{y}_{sc(n)\setminus c} \mid \mathcal{Q}_{-c}, \mathcal{X})$$

If n is a Leaf Node. If n is a leaf node $L_c^{(i)}$, its scope is $\mathbf{y}_c$ (Property 1). By the definition of a probability distribution, integrating over its entire domain yields 1:

$$\int p_L^{(i)}(\mathbf{y}_c \mid \mathcal{Q}_c, \mathcal{X})\, d\mathbf{y}_c = 1$$

This echos with the principle of PC that the marginalization of a node is by replacing its distribution with 1. The resulting 1 is the identity element for product nodes, effectively *removing* the channel from the joint factorization.

If $sc(n) \cap \mathbf{y}_c = \emptyset$, the leaf is independent of the marginalization variable, and is therefore not affected.

If n is a Product Node. By decomposability, y_c exists in the scope of exactly one child, say child k'.

$$p_n(\mathbf{y}_{sc(n)} \mid \mathcal{Q}, \mathcal{X}) = p_{k'}(\mathbf{y}_{sc(k')} \mid \mathcal{Q}, \mathcal{X}) \cdot \prod_{k \neq k'} p_k(\mathbf{y}_{sc(k)} \mid \mathcal{Q}, \mathcal{X})$$

Integrating over $\mathbf{y}_c$:

$$\int \left(p_{k'}(\mathbf{y}_{c(k')} \mid \mathcal{Q}, \mathcal{X}) \prod_{k \neq k'} p_k(\mathbf{y}_{sc(k)} \mid \mathcal{Q}, \mathcal{X}) \right) d\mathbf{y}_c = \left(\int p_{k'}(\mathbf{y}_{sc(k')} \mid \mathcal{Q}, \mathcal{X}) d\mathbf{y}_c \right) \prod_{k \neq k'} p_k(\mathbf{y}_{sc(k)} \mid \mathcal{Q}, \mathcal{X})$$

If $sc(k') = \mathbf{y}_c$, the integral is 1, effectively removing channel c from the product.

If n is a Sum Node. By Smoothness (see Section 3.2), all children share the same scope. By Property 3, weights $w_{n,k}$ are independent of $\mathbf{y}$ and $\mathcal{Q}$.

$$\int \left(\sum_{k \in ch(n)} w_{n,k}(\mathcal{X}) \cdot p_k(\mathbf{y}_{sc(n)} \mid \dots) \right) d\mathbf{y}_c = \sum_{k \in ch(n)} w_{n,k}(\mathcal{X}) \int p_k(\mathbf{y}_{sc(n)} \mid \dots) d\mathbf{y}_c$$

Each integral $\int p_k(\mathbf{y}_{sc(n)}) d\mathbf{y}_c$ yields the marginal $p_k(\mathbf{y}_{sc(n)\setminus c})$. Since the weights $w_{n,k}(\mathcal{X})$ remain unchanged, the node n now represents a valid mixture of the marginalized distributions of its children.

Since the property holds for the leaves and is preserved by both product and sum operations, it holds for the root of the circuit. By repeating this process for any number of channels $c \in \{1, \dots, C\}$, we prove that the PC is marginalization consistent across all channel-level subsets. □

A.2. CircuITS's Leaves are Marginalization Consistent

Proposition A.2 (Marginalization Consistency of CircuITS' Leaf Distributions)**.** *Let $\mathcal{Q}_c \subseteq \mathcal{Q}$ be the subset of forecasting queries belonging to a specific channel c, and let $\mathbf{y}_c$ be the vector of target variables corresponding to $\mathcal{Q}_c$.*

Consider an arbitrary query index m within this channel's set $\mathcal{Q}_c$. Let $\mathcal{Q}_{c\setminus m}$ denote the query set with the m-th query point removed, and $\mathbf{y}_{c\setminus m}$ denote the target vector without y_m. The leaf distribution for channel c is marginalization consistent if:

$$\int p(\mathbf{y}_c \mid \mathcal{Q}_c, \mathcal{X})\, dy_m = p(\mathbf{y}_{c\setminus m} \mid \mathcal{Q}_{c\setminus m}, \mathcal{X})$$

By induction, satisfying this condition implies consistency for any subset of variables within the channel.

Proof. The leaf distribution for a channel c and mixture component k is defined as a Gaussian Copula combined with univariate Deep Sigmoidal Flows (DSF) (Nelsen, 2006). It is a standard property of the Gaussian Copula that marginalizing out a single variable y_m yields a Gaussian Copula over the remaining variables $\mathbf{y}_{c\setminus m}$, parameterized by the correlation matrix $R^{c,k}$ with the m-th row and column removed.

Therefore, validity relies on **parameter stability**: the model parameters for the remaining queries $\mathcal{Q}_{c\setminus m}$ must not change when the query point q_m is removed. Specifically, for any two remaining indices $i, j \neq m$, the correlation entry R_{ij} must remain identical.

In CircuITS, the correlation matrix entry for channel c and component k is computed as:

$$R_{ij} = \frac{\mathbf{v}_i^\top \mathbf{v}_j}{H} + \delta_{ij}(1 - \dots),$$

where $\mathbf{v}_i = \tanh(\text{MLP}(e_i))$. The context embedding e_i is derived via $e_i = \text{Proj}([\phi(t_i^{\text{qry}})||\tilde{h}_c])$ (see Equations (14) and (21) to (23)).

Crucially, the channel context $\tilde{h}_c$ is computed solely from the observation history $\mathcal{X}$ via the set-based encoder and is independent of the query set $\mathcal{Q}_c$. Thus, e_i and $\mathbf{v}_i$ depend strictly on the specific query point $q_i = (t_i^{\text{qry}}, c)$ and $\mathcal{X}$. Since the calculation of R_{ij} for $i, j \neq m$ does not involve the query point q_m, removing q_m simply corresponds to removing the m-th row and column from $R^{c,k}$. This matches the theoretical marginalization requirement of the Gaussian Copula. □

A.3. Proof of CircuITS' Marginalization Consistency

CircuITS is a marginalization consistent model for conditional joint density estimation of irregular multivariate time series forecasting. Formally, for any subset of variables $\mathbf{y}_A \subset \mathbf{y}$:

$$\int p(\mathbf{y} \mid \mathcal{Q}, \mathcal{X})\, d\mathbf{y}_A = p(\mathbf{y} \setminus \mathbf{y}_A \mid \mathcal{Q} \setminus \mathcal{Q}_A, \mathcal{X})$$

Proof. Let S be an arbitrary set of indices, and let $S_c = S \cap \mathcal{I}_c$ denote the indices belonging to channel c, such that $S = \bigcup_{c=1}^{C} S_c$.

From Lemma A.1, we have established that the circuit structure is marginalization-consistent up to the leaf nodes. The integral operator distributes through the sum and product nodes such that the global integral is determined by the local integrals at the leaves:

$$\int p(\mathbf{y} \mid \mathcal{Q}, \mathcal{X}) d\mathbf{y}_{S_c} = \text{PC}\left(\dots, \int p_L^{(i)}(\mathbf{y}_c \mid \mathcal{Q}_c, \mathcal{X}) d\mathbf{y}_{S_c}, \dots\right)$$

where $\text{PC}(\cdot)$ represents the original circuit topology with fixed weights $w_{n,k}(\mathcal{X})$ (Property 3 from Lemma A.1).

By Proposition A.2, each leaf node is itself marginalization consistent over its subset S_c:

$$\int p_L^{(i)}(\mathbf{y}_c \mid \mathcal{Q}_c, \mathcal{X}) d\mathbf{y}_{S_c} = p_L^{(i)}(\mathbf{y}_{c\setminus S_c} \mid \mathcal{Q}_{c\setminus S_c}, \mathcal{X})$$

Substituting this into the circuit, the leaf nodes are replaced by their corresponding marginal distributions over the remaining queries $\mathcal{Q}_{c\setminus S_c}$. By repeating this process for all c, the entire circuit represents the valid marginal distribution $p(\mathbf{y}_{-S} \mid \mathcal{Q}_{-S}, \mathcal{X})$. □

B. Architecture diagram of CircuITS' Encoder

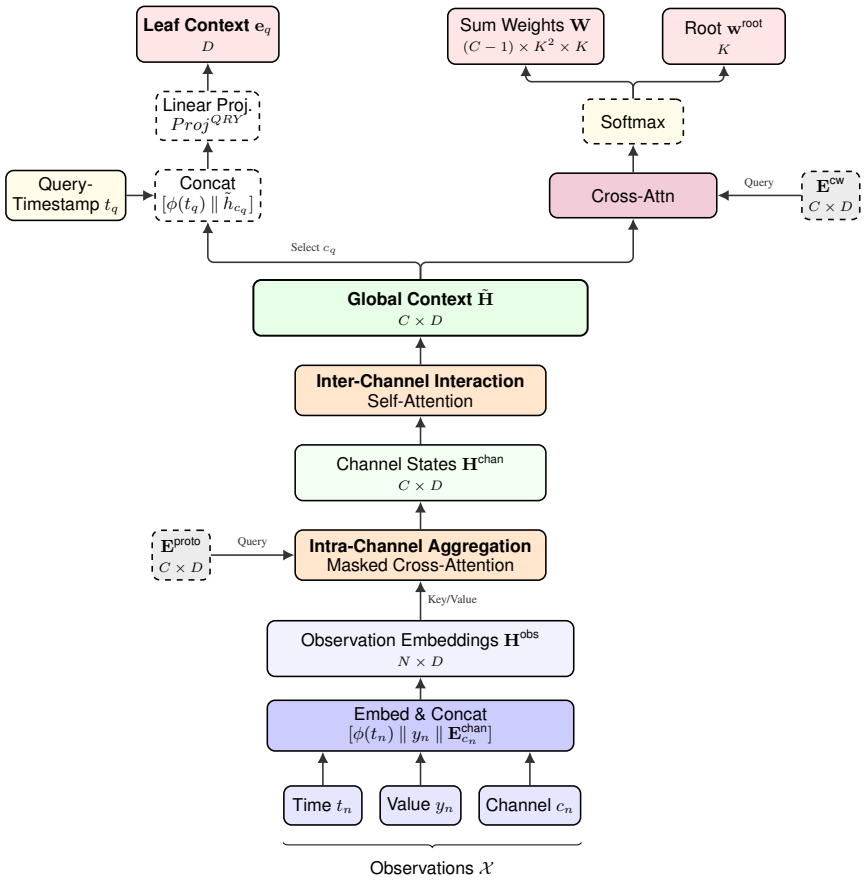


Figure 3. Detailed architecture of the CircuITS Encoder with tensor dimensions. The global context $\tilde{\mathbf{H}}$ (dim $C \times D$) branches to generate leaf contexts (dim D) and circuit weights.

C. Sampling with CircuITS' Sum Product Network

Let $z_c \in \{1, \ldots, K\}$ denote the index of the active sum node (latent component) for channel c. The sampling proceeds as follows:

Root Initialization ($c = C$): We begin by sampling the final latent state z_C directly from the distribution defined by the final weights:

$$z_C \sim \text{Categorical}(\mathbf{w}^{\text{root}}) \tag{27}$$

Recursive Backtracking ($c = C, \ldots, 2$): Given the current active state $z_c = k$, we must determine the branch (product node) that generated it. The sum node k aggregates over K^2 product terms formed by the previous channel state and the current channel leaf. We sample the flattened product index $m \in \{1, \ldots, K^2\}$ using the mixing weights $W^{(c)}$ specific to the active node k:

$$m \sim \text{Categorical}(W^{(c)}_{\cdot,k}) \tag{28}$$

The sampled index m uniquely maps to a pair of indices: the latent state of the previous channel z_{c-1} and the specific mixture component j_c for the current channel's leaf distribution. We update the active state for the next step (z_{c-1}) and record the active leaf component index j_c for the current channel.

Termination ($c = 1$): At the final step of the backward pass, the remaining active state z_1 directly indicates the active leaf component j_1 for the first channel.

D. Parallelizing the Circuit Computation

The recursive Sum-Product Network (SPN) aggregation described in Section 4.1 inherently suggests a sequential computation: the joint state $\phi^{(c)}$ depends on $\phi^{(c-1)}$, requiring $O(C)$ sequential steps for C channels. However, by analyzing the algebraic structure of the update rule, we can reformulate the recurrence as a sequence of matrix multiplications in the log-semiring. This reformulation allows us to utilize parallel prefix scan algorithms (associative scans), reducing the time complexity from linear $O(C)$ to logarithmic $O(\log C)$ on parallel hardware (e.g., GPUs).

D.1. Theoretical Background: Connection to HMMs

The recursive update in CircuITS is isomorphic to the Forward Algorithm in a Hidden Markov Model (Rabiner, 1989) (HMM) with a time-varying transition matrix.

Recall the update rule for the circuit state. Let $\log \phi^{(c-1)} \in \mathbb{R}^K$ be the log-density vector of the joint state after $c-1$ channels. The update for the k-th component of the next channel c is given by aggregating over the previous states i and the current leaf mixture components j:

$$\log \phi_k^{(c)} = \log \sum_{i=1}^{K} \sum_{j=1}^{K} \exp\left(\log \phi_i^{(c-1)} + L_{c,j} + \log W_{(i,j),k}^{(c)}\right)$$

where $L_{c,j}$ is the log-likelihood of the j-th leaf component for channel c, and $W^{(c)}$ describes the mixing weights. We can rearrange the summation to isolate the dependence on the previous state $\phi_i^{(c-1)}$:

$$\log \phi_k^{(c)} = \log \sum_{i=1}^{K} \exp\left(\log \phi_i^{(c-1)} + \underbrace{\log \sum_{j=1}^{K} \exp\left(L_{c,j} + \log W_{(i,j),k}^{(c)}\right)}_{M_{ik}^{(c)}}\right)$$

Here, we define a transition matrix $M^{(c)} \in \mathbb{R}^{K \times K}$ for step c. The entry $M_{ik}^{(c)}$ represents the log-probability of transitioning from joint component i to component k, marginalizing over the leaf distributions of the current channel.

In vector notation, this becomes a matrix-vector multiplication in the log-semiring (where $(\oplus, \otimes) \to (\text{logsumexp}, +)$):

$$\phi^{(c)} = \phi^{(c-1)} \otimes_{\log} M^{(c)}$$

Consequently, the final joint distribution weights $\phi^{(C)}$ can be expressed as the product of the initial state $\phi^{(1)}$ and a sequence of transition matrices:

$$\phi^{(C)} = \phi^{(1)} \otimes_{\log} M^{(2)} \otimes_{\log} M^{(3)} \cdots \otimes_{\log} M^{(C)}$$

This formulation highlights that CircuITS, while structurally an SPN, shares the computational characteristics of linear recurrent models and HMMs (Särkkä & García-Fernández, 2021).

D.2. Parallel Log-Semiring Scan

Since matrix multiplication is associative, the cumulative product of the transition matrices $M^{(c)}$ can be computed in parallel using the Hillis-Steele algorithm (Hillis & Steele, 1986).

Instead of computing the product sequentially $(((M^{(2)}M^{(3)})M^{(4)})\dots)$, a parallel scan computes the products in a tree-like structure. For a sequence of length C, this requires $\mathcal{O}(\log C)$ sequential steps with sufficient parallel threads.

E. Details on the Synthetic Bifurcation Experiment

For the toy experiment described in Section 5, we generated IMTS trajectories using 50 evenly spaced time steps. The data points prior to bifurcation are randomly distributed around 0 with a standard deviation of 0.1. After the bifurcation point (at the 18th time step), the channels follow a Brownian motion with a drift of either 0.1 or -0.1 per time step. The standard deviation is still 0.1. This decision is random, independent per channel, and equiprobable for both paths.

The models are tasked with predicting the distribution of the final 75% of time steps, conditioned on the initial 25%. Consequently, the observations provided as input contain no information regarding the future drift direction. To simulate irregularity, we randomly dropped 5% of the observations and query points.

We present samples from ProFITi in Figure 4, along with the final normalized joint negative log-likelihood (njNLL) on the test split for ProFITi, MOSES, and CircuITS. While ProFITi correctly captures the Brownian motion dynamics and approaches CircuITS in modeling the joint likelihood (Table 3), it fails to learn the strict separation of modalities.

Table 3. Test njNLL for the synthetic four channel bifurcation task.

	MOSES-10	MOSES-16	ProFITi	CircuITS
njNLL	-0.340 ±.07	-0.290 ± .04	-1.253 ± .01	**-1.280 ± .02**

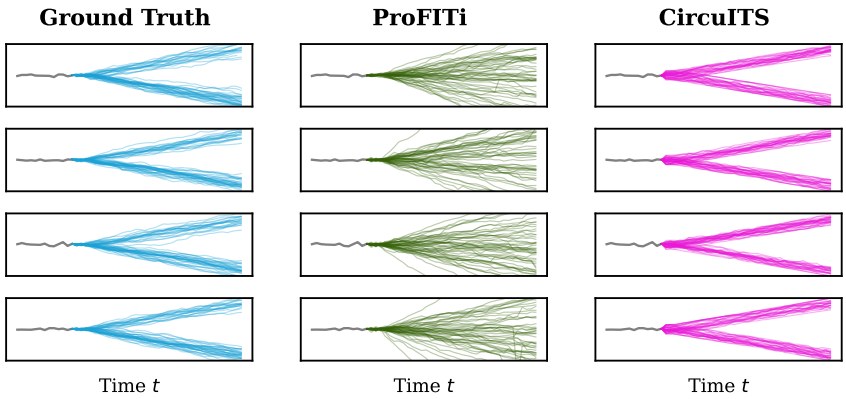


Figure 4. Comparing samples from ProFITi and CircuITS for the four channel bifurcation task.

F. Dataset Descriptions

In coherence with Yalavarthi et al. (2025b) we utilize the following four benchmark datasets to evaluate performance, covering both climatological and physiological domains. The specific characteristics and preprocessing protocols for each are detailed below:

USHCN. Derived from the U.S. Historical Climatology Network (Menne et al., 2006), this dataset tracks climate changes across the USA. It consists of daily measurements of 5 variables (snow precipitation, rain precipitation, snow depth, min. temperature, and max temperature) collected over 150 years from 1,218 meteorological stations. Following the protocol of De Brouwer et al. (2019), we selected a subset of 1,100 stations and an observation window of 4 years (1996-2000). USHCN is transformed into an IMTS by randomly dropping 95% recorded observations.

PhysioNet2012. Sourced from the PhysioNet Computing in Cardiology Challenge 2012 (Silva et al., 2012), this physiological dataset contains medical records of 12,000 patients admitted to the ICU. It tracks 37 vital signs over a 48-hour period. Consistent with Che et al. (2018), the dataset is binned into hourly observations by rounding time stamps to the hour and averaging observations, if a channel is observed multiple times within one hour.

MIMIC-III. The Medical Information Mart for Intensive Care III (Johnson et al., 2016) is a large-scale physiological dataset collected at Beth Israel Deaconess Medical Center. It contains 17,874 clinical instances measuring 96 variables over 48 hours. Following De Brouwer et al. (2019), observations are aggregated and rounded to 30-minute intervals.

MIMIC-IV. This dataset features records from a tertiary academic medical center in Boston. It tracks 102 variables from ICU patients over a 48-hour period (Johnson et al., 2023a;b). Adopting the standards of Biloš et al. (2021), the observations are rounded to 1-minute intervals. Due to the more fine-grained binning, this dataset records the greatest number of observations and queries.

Table 4. Summary of Dataset Characteristics and Preprocessing

Dataset	Samples	Channels	Duration	Binning	Max. Observations
USHCN	1,100 (Stations)	5	4 Years	1 Week	325
PhysioNet2012	12,000 (Patients)	37	48 Hours	1 Hour	552
MIMIC-III	21,000 (Patients)	96	48 Hours	30 Mins	741
MIMIC-IV	18,000 (Patients)	102	48 Hours	1 Min	1386

Table 5. Minimum, Average, and Maximum Number of Queries (Grouped by Task)

	36-3			36-12			24-24			12-36		
Dataset	min.	avg.	max.	min.	avg.	max.	min.	avg.	max.	min.	avg.	max.
USHCN	3	3.3	6	5	54.7	63	19	109.4	120	19	164.1	179
PhysioNet2012	1	19.8	53	1	61.8	148	1	139.9	297	1	220.7	459
MIMIC-III	1	10.3	85	1	25.3	264	1	67.2	512	1	109.7	785
MIMIC-IV	1	7.8	79	1	57.2	481	1	128.9	906	1	208.4	1357

G. Hyperparameters

Each model is trained for a maximum of 2000 epochs using early stopping with patience of 30 epochs. We use the AdamW (Kingma & Ba, 2015; Loshchilov & Hutter, 2019) optimizer with an initial learning rate of 0.001, an L2 regularization of 0.001 and a batch size of 64. Additionally, we use a learning-decay-on-plateau scheduler with a decay rate of 0.5 and 5 epochs patience. For each model we evaluate a maximum of 10 randomly searched hyperparameter setting. We train each setting on single fold and select the setting with the best validation njNLL. We follow Yalavarthi et al. and select the following hyperparameter search spaces for the evaluated models:

MOSES. For the MOSES model (Yalavarthi et al., 2025b), we perform a search over the number of **mixture components** $\in \{1, 2, 5, 7, 10\}$. Additionally, we evaluate different sizes for the **hidden dimension** $\in \{16, 32, 64, 128\}$ and vary the number of **attention heads** $\in \{1, 2, 4\}$.

ProFITi. For ProFITi (Yalavarthi et al., 2025a), we evaluate the number of **flow layers** $\in \{7, 8, 9, 10\}$ and the **latent dimension size** $\in \{32, 64, 128, 256\}$.

Neural Flows. For the Neural Flows model (Biloš et al., 2021), we primarily search over the number of flow layers $\in \{1, 4\}$. We fix the **hidden dimension** to 64, and the number of **hidden layers** is 2, and the **flow model** is set to GRU.

CircuITS. For our model, we vary the number of **(components** K**)** $\in \{2, 3, 4\}$ and the **hidden dimension** $D' \in \{32, 64\}$. For the Deep sigmoidal flow on the univariate marginals we tune the number of **flow layers** $\in \{2, 3\}$, and the **flow hidden dimension** $\in \{10, 20\}$. The MLP that infers the parameters has 2 hidden layers and a hidden dimension of 32. The multi-head attention layers in the encoder are applied by setting the number of **attention heads** to 2. For CircuITS we had to set the batch size to 32 for the MIMIC-IV due to memory overflow.

The remaining hyperparameters of the baseline models are taken from the respective papers.

H. Details on Ablation Studies

We provide additional information for the ablation studies presented in Table 2.

w/o SPN. We conduct this experiment by training a model where the number of leaf components K is set to 1. Consequently, there is no weighted sum of different components; instead, channels are combined via a single product distribution.

w/o DSF. We replace the Deep Sigmoidal Flow (DSF), which is applied to the univariate marginals, with normal distributions parameterized by (μ, σ) that are inferred from the embeddings of the forecasting queries. Formally:

$$\mu_i^k = f^{\text{mean}}(e_i^k) \tag{29}$$

$$\sigma_i^k = \exp(f^{\text{var}}(e_i^k)), \tag{30}$$

where f^{mean} and f^{var} are implemented as 2-layer MLPs. We do not change the computation of the correlation matrix R.

w/o GC. For this ablation, we set the correlation matrix $R = I$, thereby treating the univariate marginals of a single component k as independent.

w/ TF. We utilize Tripletformer (TF) (Yalavarthi et al., 2023) as an encoder to compute the circuit weights and the embeddings of the forecasting queries. In contrast to the encoder in CircuITS, Tripletformer does not group observations/queries by channel. Instead, it first performs self-attention among all observation triplets (channel, timestamp, value) and subsequently applies cross-attention from the encoded observation triplets to the forecasting queries.

To infer the weights of the SPN, we include an additional Cross-Attention layer with learnable circuit weight embeddings $\mathbf{E}^{\text{cw}}$ (similar to Equation (15), but using all encoded observation triplets as attention keys and values).

I. Additional Evaluation

I.1. More Baselines and MSE of Means

In addition to the baselines presented in Section 5, we compare CircuITS to some additional baselines. Similar to NeuralFlows (Biloš et al., 2021), Continuous Recurrent Units (CRU) (Schirmer et al., 2022) and GRU-ODE-Bayes (De Brouwer et al., 2019) learn to forecasting univariate marginals of an IMTS.

Furthermore, we compare with Gaussian Process Regression (Dürichen et al., 2015) and Gaussian Mixture Model (Yalavarthi et al., 2025b), which predicts a mixture of multivariate Gaussian in the same way that MOSES (Yalavarthi et al., 2025b), however without applying normalizing flows. The njNLL and mnLL are shown in Table 6 and Table 7 shows the Mean Squared Error (MSE) of the empirical means calculated with 1000 samples.

Table 6. Comparison of njNLL and mNLL on the 36-3* task. Lower values are better. Best results are in **bold**, second best in *italics*. Results are taken from (Yalavarthi et al., 2025b). OOM stands for out-of-memory.

Model	USHCN		PhysioNet'12		MIMIC-III		MIMIC-IV	
	njNLL	mNLL	njNLL	mNLL	njNLL	mNLL	njNLL	mNLL
ProFITi	-3.226 ± .23	*-3.324 ± .21*	**-0.647 ± .08**	*-0.016 ± .09*	*-0.377 ± .03*	0.408 ± .03	*-1.777 ± .07*	*0.500 ± .32*
GRU-ODE	0.766 ± .16	0.776 ± .17	0.501 ± .00	0.504 ± .06	0.961 ± .06	0.839 ± .03	0.823 ± .32	0.876 ± .59
NeuralFlows	0.775 ± .15	0.775 ± .18	0.496 ± .00	0.492 ± .03	0.998 ± .18	0.866 ± .10	0.689 ± .09	0.796 ± .05
CRU	0.761 ± .19	0.762 ± .18	1.057 ± .01	0.931 ± .02	1.234 ± .08	1.209 ± .04	OOM	OOM
GPR	2.011 ± 1.4	1.235 ± .10	1.367 ± .07	1.161 ± .07	3.146 ± .36	1.341 ± .01	2.789 ± .05	1.161 ± .01
GMM	1.050 ± .03	1.042 ± .02	1.063 ± .00	1.069 ± .00	1.160 ± .02	1.124 ± .01	1.076 ± .00	1.075 ± .01
MOSES	*-3.357 ± .18*	-3.355 ± .16	-0.491 ± .04	-0.271 ± .03	-0.305 ± .03	*0.163 ± .03*	-1.668 ± .10	-0.634 ± .02
CircuITS	**-3.789 ± .22**	**-3.717 ± .20**	*-0.550 ± .01*	**-0.287 ± .04**	**-0.574 ± .08**	**0.095 ± .11**	**-2.113 ± .04**	**-0.731 ± .06**

Table 7. Comparing the MSE of the mean estimated by 1000 samples from probabilistic models on the 36-3* setting.

Model	USHCN	PhysioNet'12	MIMIC-III	MIMIC-IV
ProFITi	0.308 ± .06	0.305 ± .01	0.548 ± .06	0.389 ± .02
GRU-ODE	0.410 ± .11	0.329 ± .00	*0.479 ± .04*	0.365 ± .01
NeuralFlows	0.424 ± .11	0.331 ± .01	*0.479 ± .05*	0.374 ± .02
CRU	**0.290 ± .06**	0.475 ± .02	0.725 ± .04	OOM
GPR	0.597 ± .11	0.575 ± .06	0.862 ± .02	0.609 ± .01
GMM	*0.294 ± .08*	*0.293 ± .01*	0.535 ± .06	*0.332 ± .02*
MOSES	0.411 ± .10	0.307 ± .01	0.517 ± .06	0.342 ± .03
CircuITS	0.306 ± .03	**0.292 ± .00**	**0.475 ± .05**	**0.285 ± .00**

I.2. Assessing the Importance of Channel Ordering

To quantify the importance of channel ordering, we repeated the Physionet-12 evaluation (36-to-12 channels) using five randomly sampled channel permutations. The resulting test njNLL values are provided in Appendix I.2. While the results indicate that ordering has a statistically discernible impact, the variation is negligible relative to the performance gaps between the competing models. Consequently, the choice of permutation does not alter the qualitative conclusions of the experiment.

Table 8. Permuting the channels order in the recursive SPN aggregation for the Physionet'12 36–12 task.

	Original	Perm. 1	Perm. 2	Perm. 3	Perm. 4
njNLL	-0.536 ± .04	-0.540 ± .02	-0.574 ± .03	-0.526 ± .02	-0.547 ± .03

I.3. Energy Score and CRPS

To evaluate the quality of samples produced by the learned distributions, we employ the Energy Score and the Continuous Ranked Probability Score (CRPS). Since closed-form solutions are often intractable for complex joint distributions modeled by CircuITS and the baselines, we approximate these metrics using Monte Carlo estimation with $S = 1000$ samples.

Energy Score. The Energy Score is a multivariate strictly proper scoring rule that assesses the quality of the joint distribution, capturing correlations between variables in the IMTS. For a ground truth vector $y \in \mathbb{R}^M$ corresponding to the query $\mathcal{Q}$, and S samples, the empirical Energy Score is defined as:

$$ES(y, \mathcal{Q}) = \frac{1}{S}\sum_{s=1}^{S} \|\hat{y}^{(s)} - y\|_2 - \frac{1}{2S^2}\sum_{s=1}^{S}\sum_{j=1}^{S} \|\hat{y}^{(s)} - \hat{y}^{(j)}\|_2 \tag{31}$$

where $\|\cdot\|_2$ denotes the Euclidean norm. The first term measures the distance between the samples and the ground truth (calibration), while the second term measures the diversity of the samples (sharpness).

Continuous Ranked Probability Score (CRPS). While the Energy Score evaluates the joint distribution, the CRPS evaluates the accuracy of the univariate marginal distributions. We report the mean CRPS averaged over all query targets in $\mathcal{Q}$. The sample-based approximation for the CRPS is given by:

$$CRPS(y, \mathcal{Q}) = \frac{1}{|\mathcal{Q}|}\sum_{i=1}^{|\mathcal{Q}|}\left(\frac{1}{S}\sum_{s=1}^{S}|\hat{y}_i^{(s)} - y_i| - \frac{1}{2S^2}\sum_{s=1}^{S}\sum_{j=1}^{S}|\hat{y}_i^{(s)} - \hat{y}_i^{(j)}|\right) \tag{32}$$

where y_i is the ground truth value for the i-th query target and $\hat{y}_i^{(s)}$ is the corresponding value in the s-th sample. Table 9 compares these scores for NeuralFlows, ProFITi, MOSES, and CircuITS.

Table 9. Energy Score and CRPS of competing models. Lower values are better. Best results are in **bold**, second best in *italics*. Models are trained and tuned on njNLL.

	Model	USHCN		PhysioNet'12		MIMIC-III		MIMIC-IV	
		Energy	CRPS	Energy	CRPS	Energy	CRPS	Energy	CRPS
36–3*	NeuralFlows	0.661 ± .06	0.306 ± .03	1.691 ± .00	0.277 ± .00	*1.381 ± .03*	0.308 ± .00	0.982 ± .01	0.281 ± .00
	ProFITi	**0.452 ± .04**	**0.182 ± .01**	**0.879 ± .30**	0.271 ± .00	1.606 ± .17	0.319 ± .00	**0.808 ± .00**	0.279 ± .01
	MOSES	0.552 ± .04	0.220 ± .02	*1.599 ± .01*	*0.260 ± .00*	**1.353 ± .03**	*0.296 ± .01*	*0.906 ± .03*	*0.245 ± .01*
	CircuITS	*0.468 ± .02*	**0.182 ± .01**	1.613 ± .00	**0.252 ± .00**	1.489 ± .04	**0.286 ± .01**	0.927 ± .05	**0.221 ± .00**
36–12	NeuralFlows	5.943 ± .39	0.428 ± .02	6.107 ± .18	0.320 ± .00	8.107 ± .83	0.384 ± .02	12.222 ± .39	0.318 ± .01
	ProFITi	*4.385 ± 1.7*	*0.234 ± .04*	3.575 ± .45	0.312 ± .03	2.546 ± .13	0.383 ± .03	5.394 ± 155	0.654 ± .08
	MOSES	**3.252 ± .13**	**0.211 ± .01**	**3.195 ± .04**	**0.285 ± .00**	*2.358 ± .10*	*0.350 ± .01*	*2.869 ± .04*	*0.262 ± .01*
	CircuITS	5.535 ± .93	0.359 ± .06	*3.213 ± .00*	*0.286 ± .00*	**2.279 ± .01**	**0.327 ± .01**	**2.835 ± .09**	**0.249 ± .01**
24–24	NeuralFlows	*8.448 ± .66*	0.433 ± .02	9.522 ± .44	0.352 ± .00	10.290 ± .35	0.407 ± .01	16.836 ± .71	0.333 ± .01
	ProFITi	14.42 ± 7.6	*0.355 ± .06*	5.742 ± .28	0.336 ± .01	4.287 ± .38	0.438 ± .07	10.82 ± 1.1	1.132 ± .17
	MOSES	**5.330 ± .31**	**0.239 ± .01**	*5.284 ± .03*	*0.318 ± .00*	*3.632 ± .05*	*0.333 ± .00*	**4.268 ± .07**	**0.279 ± .01**
	CircuITS	10.71 ± 1.9	0.513 ± .11	**5.267 ± .01**	**0.315 ± .00**	**3.543 ± .04**	**0.323 ± .01**	*4.482 ± .15*	*0.281 ± .01*
12–36	NeuralFlows	**10.79 ± 1.1**	**0.437 ± .02**	12.170 ± .41	0.396 ± .01	14.524 ± .95	0.477 ± .00	23.049 ± .97	0.356 ± .00
	ProFITi	41.61 ± 39	0.593 ± .36	8.439 ± .42	0.408 ± .03	6.567 ± .53	2.413 ± 1.0	11.10 ± .91	0.823 ± .12
	MOSES	21.22 ± 17	1.198 ± 1.15	*7.685 ± .63*	*0.379 ± .04*	*5.770 ± .03*	*0.418 ± .00*	*6.276 ± .03*	*0.332 ± .00*
	CircuITS	*12.18 ± 1.8*	*0.456 ± .07*	**7.241 ± .01**	**0.355 ± .00**	**5.730 ± .06**	**0.409 ± .00**	**5.851 ± .05**	**0.292 ± .00**

I.4. Sensitivity Analysis for the Number of Components

We evaluated the sensitivity of CircuITS to the hyperparameter K, representing the number of leaf nodes per channel (mixture components). As shown in Figure 5, the model exhibits moderate robustness to variations in K. Our results suggest that selecting K from the set $\{2, 3, 4\}$ is sufficient, as the optimal configuration was found to be $K = 2$ or $K = 3$ for the evaluated task (36–12).

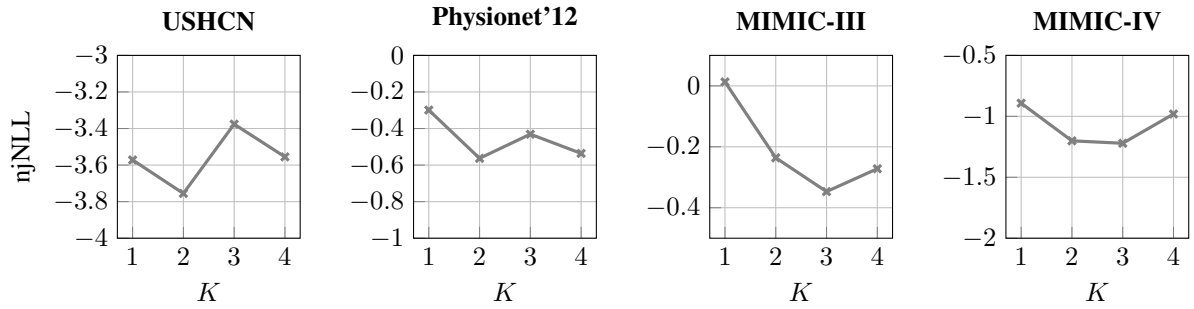


Figure 5. Sensitivity Analysis on the 36-12 task for the number of circuit components K.

J. Memory Consumption and Epoch Times

We conduct an experiment to compare CircuITS' memory consumption during training and epoch times with those of ProFITi and MOSES. To do this, we use toy datasets based on Brownian motion, in which we can vary the number of channels C and the number of time steps N. The models are tasked with predicting the second half of time steps based on the first half, similar to the 24–24 task in our main experiment (Section 5). To simulate an IMTS, we randomly drop 10% of all observations and forecasting queries. Hence, the expected number of query points $|\mathcal{Q}|$ is $\frac{0.9NC}{2}$.

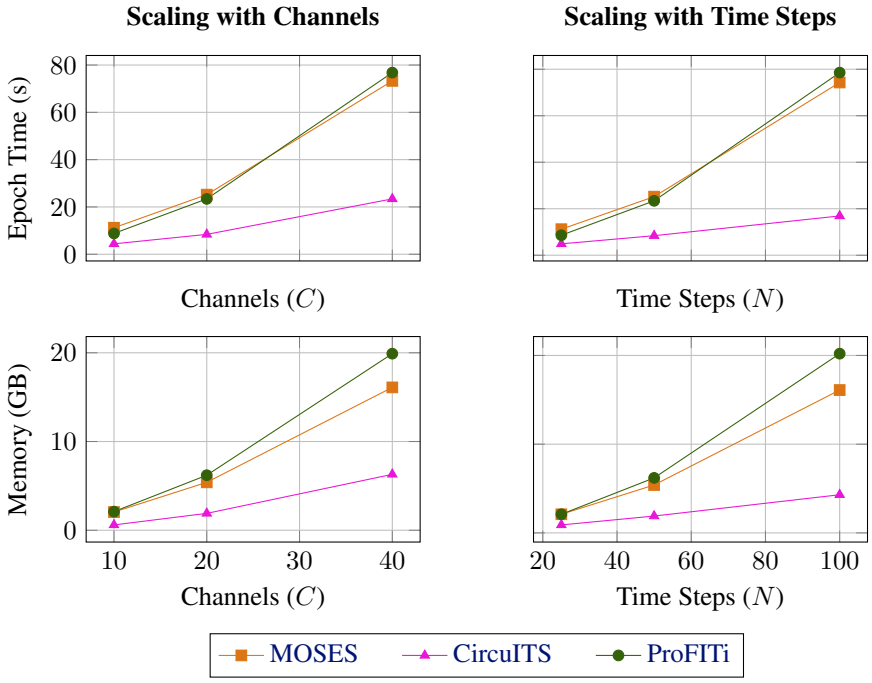


Figure 6. Comparing memory consumption and epoch times during training on Toy datasets.

For MOSES, we set the number of mixture components to 3, the hidden dimension to 128, and the number of attention heads to 4. ProFITi is applied using 7 flow layers and a latent dimension of 64. For CircuITS, we set the number of components K

to 3, the number of flow layers to 2, and the hidden dimension to 32. We run these experiments on a single NVIDIA A40 with a batch size of 64. Figure 6 presents the results. In this toy setup, where the channels all have a similar number of queries, CircuITS is significantly faster than MOSES and ProFITi, while using less memory to compute the gradients and update the weights.

As stated in Section 4.5 CircuITS reduces the complexity compared to a model that computes full covariance over all query points by factor C^2 under the assumption that the forecasting targets are distributed evenly among channels.

However, in real world IMTS datasets this is typically not the case and some channels are observed more frequently than others. At least in our implementation the speed-up that our model generates by grouping the forecasting targets by channels decreases, when forecasting queries are concentrating on single channels. As an example we show the distribution of queries over channels for MIMIC-III in the 12–36 setting in Figure 7. For such uneven IMTS CircuITS is actually slower than MOSES and ProFITi as we show in Table 10.

Table 10. Epoch times on MIMIC-III. The Experiment were run on a single NVIDIA A100 GPU

Model	36–12	24–24	12–36
ProFITi	13s	14s	15s
MOSES	13s	12s	13s
CircuITS	27s	24s	40s

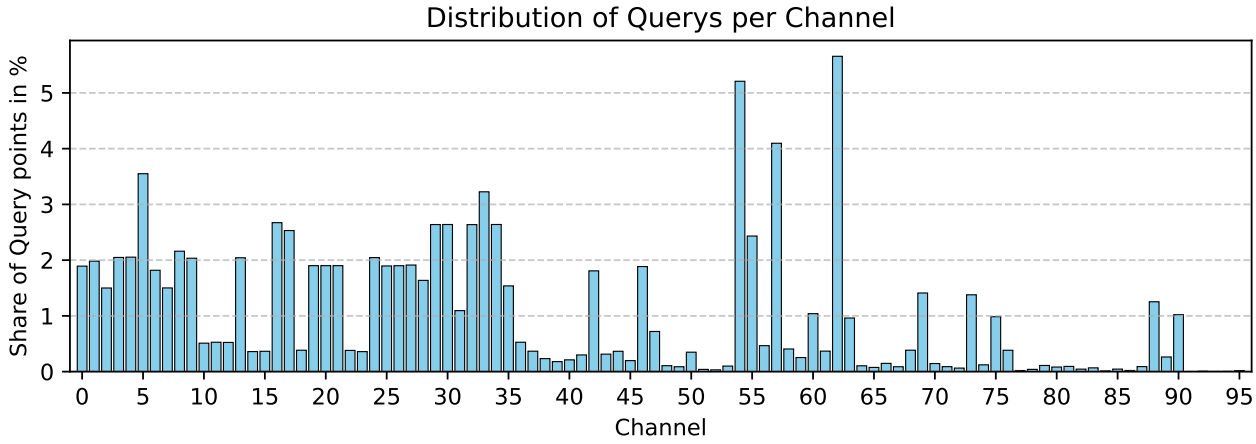


Figure 7. Query distribution for MIMIC-III 12–36